Examination Questions

ELEC50002: Communications, April 2024
Dept. of Electrical and Electronic Engineering, Imperial College London

Name: Solution CID: _____

Instruction:

- This is a closed-book examination. But a standard calculator and three pages (8½" X11") of handwriting notes are allowed.
  - *Standard calculator*: please check the EEE Dept, Imperial College London, for the specifications of a standard calculator.
  - *Handwriting notes*: the notes must be in your original handwriting on three pieces of standard letter size paper (one side only). Please write your name and CID on the top left on each page of the notes.
- Incorrect answers with justification and reasoning can receive partial credit. However, correct answer without justification or reasoning will receive no credit.
- Students need to answer all 6 problems.

Prob I (*10 pts*)	
Prob II (*10 pts*)	
Prob III (*8 pts*)	
Prob IV (*11 pts*)	
Prob V (*9 pts*)	
Prob VI (*12 pts*)	
Total (*60 pts*)	

Some Fourier Transform Pairs

	$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$	$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$
1	$x(t) = 1$	$X(f) = \delta(f)$
2	$x(t) = \delta(t)$	$X(f) = 1$
3	$x(t) = u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$
4	$x(t) = sgn(t) = \begin{cases} 1, & t \geq 0 \\ -1, & t < 0 \end{cases}$	$X(f) = \frac{1}{j\pi f}$
5	$x(t) = \frac{1}{\pi t}$	$X(f) = -jsgn(f) = \begin{cases} -j, & f \geq 0 \\ j, & f < 0 \end{cases}$
6	$x(t) = rect\left(\frac{t}{T}\right) = \begin{cases} 1, & \lvert t\rvert \leq T/2 \\ 0, & \lvert t\rvert > T/2 \end{cases}$	$X(f) = T\frac{\sin(\pi fT)}{\pi fT} = Tsinc(fT)$
7	$x(t) = W\frac{\sin(\pi Wt)}{\pi Wt} = Wsinc(Wt)$	$X(f) = rect\left(\frac{f}{W}\right) = \begin{cases} 1, & \lvert f\rvert \leq W/2 \\ 0, & \lvert f\rvert > W/2 \end{cases}$
8	$x(t)\cos(2\pi f_o t)$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
9	$x(t)\sin(2\pi f_o t)$	$\frac{1}{2j}X(f - f_0) - \frac{1}{2j}X(f + f_0)$

Fourier Transform Properties

	Name of Theorem	Function	Fourier Transform
1	Superposition (a_1 and a_2 are arbitrary constants)	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(f) + a_2X_2(f)$
2	Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
3	Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
4	Time reversal	$x(-t)$	$X(-f) = X^*(f)$, * is complex conjugate
5	Duality	$X(t)$	$x(-f)$
6	Frequency translation	$x(t)e^{j2\pi f_o t}$	$X(f - f_0)$
7	Modulation	$x(t)\cos(2\pi f_0 t)$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
8	Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j2\pi f)^n X(f)$

9	Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$(j2\pi f)^{-1}X(f)+\frac{1}{2}X(0)\delta(f)$
10	Convolution	$\int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$ $=\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau$	$X_1(f)X_2(f)$
11	Multiplication	$x_1(t)x_2(t)$	$\int_{-\infty}^{\infty} X_1(f-\xi)X_2(\xi)d\xi$ $=\int_{-\infty}^{\infty} X_1(\xi)X_2(f-\xi)d\xi$

Some Trigonometry Formulas

$\sin(x+y)=\sin x\cos y+\cos x\sin y$	$\sin(x-y)=\sin x\cos y-\cos x\sin y$
$\cos(x+y)=\cos x\cos y-\sin x\sin y$	$\cos(x-y)=\cos x\cos y+\sin x\sin y$
$\sin x\cos y=\frac{1}{2}[\sin(x+y)+\sin(x-y)]$	$\cos x\sin y=\frac{1}{2}[\sin(x+y)-\sin(x-y)]$
$\sin x\sin y=\frac{1}{2}[\cos(x-y)-\cos(x+y)]$	$\cos x\cos y=\frac{1}{2}[\cos(x+y)+\cos(x-y)]$
$\sin x+\sin y=2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$	$\sin x-\sin y=2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$
$\cos x+\cos y=2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$	$\cos x-\cos y=-2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$

Q Function

$$Q(x)=\frac{1}{\sqrt{2\pi}}\int_{x}^{\infty} e^{\frac{-t^2}{2}}dt$$

Approximation of Q Function

$$Q(x)=\frac{1}{\sqrt{2\pi}}\int_{x}^{\infty} e^{\frac{-t^2}{2}}dt\approx\frac{1}{\sqrt{2\pi}x}e^{\frac{-x^2}{2}}\ \text{ for }\ x\gg 1$$

Joint Gaussian Density

The joint probability density function (pdf) of two correlated Gaussian random variables X and Y is given by

$$f_{xy}(x,y)=\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2}-2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]}$$

where $\mu_X=E[X]$, $\mu_Y=E[Y]$ are the mean values, $\sigma_X=\sqrt{E[(X-\mu_x)^2]}$ and $\sigma_Y=\sqrt{E\left[\left(Y-\mu_y\right)^2\right]}$ are the standard deviations of X and Y, respectively, and ρ is the correlation coefficient defined as $\rho=\frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$.

Problem I: (*Noise*) Consider the canonical form of passband noise, $N(t) = N_I(t)\cos(300\pi t) - N_Q(t)\sin(300\pi t)$, where $N_I(t)$ and $N_Q(t)$ are the corresponding in-phase and quadrature components and are independent and wide-sense stationary (WSS) Gaussian random processes with zero mean and the same power spectrum density. Denote the power spectrum density of $N_I(t)$ and $N_Q(t)$ as

$$S(f) = \begin{cases} \frac{1}{4}, & |f| \le 4, \\ 0, & |f| > 4. \end{cases}$$

a) [**2 points**] Find the variance of $N_I(t)$;

$$\mathrm{Var}(N_I(t)) = \int_{-\infty}^{+\infty} S(f)\,df = \int_{-4}^{4} \tfrac{1}{4}\,df = 2$$

b) [**2 points**] Find $E(N(t))$;

$$E(N(t)) = E\left(N_I(t)\right)\cos(300\pi t) - E\left(N_Q(t)\right)\sin(300\pi t)$$
$$= 0$$

c) [**2 points**] Find $E(N^2(t))$;

$$E\left(N^2(t)\right) = \mathrm{Var}\left(N(t)\right) + \left(E(N(t))\right)^2 = 2 + 0^2 = 2$$

d) [**2 points**] Find the probability density function of $N(0)$;

$N(0)$ is Gaussian with $\mu = E(N(0)) = 0$ and $\sigma^2 = 2$, then

$$f_{N(0)}(n) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{n^2}{2\sigma^2}\right\} = \frac{1}{2\sqrt{\pi}}\exp\left\{-\frac{n^2}{4}\right\}$$

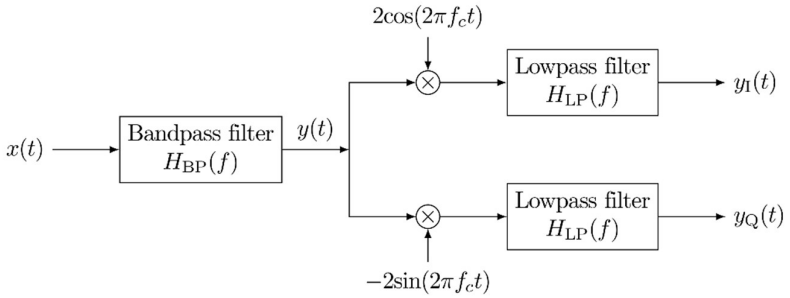
e) [**2 points**] Find $S_N(f)$, the power spectrum density of $N(t)$;

$$R_{N(t)}(\tau) = R_{N_I(t)}(\tau)\cos(2\pi f_c \tau)$$

$$S_N(f) = \frac{1}{2}S(f+f_c) + \frac{1}{2}S(f-f_c)$$

$$= \begin{cases} \frac{1}{8} & \text{if } |f-150| \le 4 \text{ or } |f+150| \le 4, \\ 0 & \text{otherwise.} \end{cases}$$

Problem II. (*Down-Conversion*) The receiver block diagram of a communication system is shown as in the following figure.



In the above figure, carrier frequency $f_c = 100$ Hz and the received signal before the passband filter is $x(t) = \cos(190\pi t) - \sin(230\pi t)$.

a) [**2 points**] Express $x(t)$ into the form, $x(t) = x_i(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)$, where $x_I(t)$ and $x_Q(t)$ are the in-phase and quadrature components corresponding $f_c = 100$ Hz.

$x(t) = \cos(200\pi t - 10\pi t) - \sin(200\pi t + 30\pi t)$

$= \cos(10\pi t)\cos(200\pi t) + \sin(10\pi t)\sin(200\pi t)$

$- \sin(200\pi t)\cos(30\pi t) - \cos(200\pi t)\sin(30\pi t)$

$= (\cos(10\pi t) - \sin(30\pi t))\cos(200\pi t) - (-\sin(10\pi t) + \cos(30\pi t))\sin(200\pi t)$

b) [**3 points**] If the frequency responses of the bandpass filter and the lowpass filters are

$$H_{\mathrm{BP}}(f) = \begin{cases} 1, & \text{if } |f-100| \le 20 \text{ Hz or } |f+100| \le 20 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases} \quad \text{and} \quad H_{\mathrm{LP}}(f) = \begin{cases} 1, & \text{if } |f| \le 20 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases}$$

find $y(t)$, $y_I(t)$, and $y_Q(t)$.

$y(t) = x(t) = (\cos(10\pi t) - \sin(30\pi t))\cos(200\pi t)$

$- (-\sin(10\pi t) + \cos(30\pi t))\sin(200\pi t)$

$y_I(t) = \cos(10\pi t) - \sin(30\pi t)$

$y_Q(t) = -\sin(10\pi t) + \cos(30\pi t)$

c) [**2 points**] Find $y_I(t)$ and $y_Q(t)$ if the frequency responses of the bandpass filter and the lowpass filters are

$$H_{\mathrm{BP}}(f) = \begin{cases} 1, & \text{if } |f-100| \le 20 \text{ Hz or } |f+100| \le 20 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases} \quad \text{and} \quad H_{\mathrm{LP}}(f) = \begin{cases} 1, & \text{if } |f| \le 10 \text{ Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

$y_I(t) = \cos(10\pi t)$

$y_Q(t) = -\sin(10\pi t)$

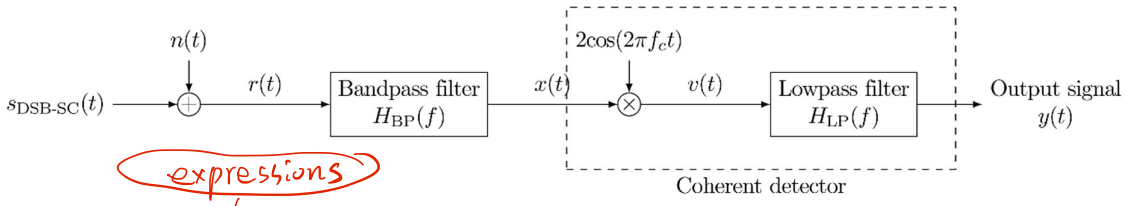
d) [**2 points**] Find $y_I(t)$ and $y_Q(t)$ if the frequency responses of the bandpass filter and the lowpass filters are

$$H_{\mathrm{BP}}(f) = \begin{cases} 1, & \text{if } |f-100| \le 10 \text{ Hz or } |f+100| \le 10 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases} \quad \text{and} \quad H_{\mathrm{LP}}(f) = \begin{cases} 1, & \text{if } |f| \le 10 \text{ Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

$y_I(t) = \cos(10\pi t)$

$y_Q(t) = -\sin(10\pi t)$

Problem III. (*Analoge Modulation*) Let $m(t)$ be a real message signal with $E[m^2(t)] = 1$, $m_P = max|m(t)| = 2$, and bandwidth $W = 15$ Hz. Denote $s_{DS}(t) = m(t)\cos(1000\,\pi t)$ as the signal of double sideband-suppressed carrier (DSB-SC) modulation.



expressions

a) [**3 points**] If the coherent detector in the above figure is used at the receiver, find carrier frequency f_c, the ~~bandwidths~~ of the bandpass filter, $H_{\text{BP}}(f)$ and the lowpass filter, $H_{\text{LP}}(f)$.

$f_c = 500$ Hz

$$H_{BP} = \begin{cases} 1, & |f| \le 15 \\ 0, & \text{otherwise} \end{cases}$$

$$H_{LP} = \begin{cases} 1, \text{ if } |f-f_c| \le 15 \text{ or } |f+f_c| \le 15 \\ 0, \text{ otherwise} \end{cases}$$

b) [**3 points**] Find the signal-to-noise ratio (SNR) at the output of the coherent detector if the channel noise is white Gaussian with power spectral density (PSD), $\frac{N_o}{2} = 10^{-4}$ Watt/Hertz.

$$SNR = \frac{A^2 P}{2N_0 W} = \frac{1^2 \times 1}{2\times(2\times10^{-4})\times 15}$$

$$= 1000/6$$

$$= 166.7 \text{ or } 22.23 \text{ dB}$$

c) [**3 points**] In the above figure, if received signal $r(t)$ ~~in the above figure~~ is added by $3\cos(1000\,\pi t)$, then

$$\hat{r}(t) = r(t) + 3\cos(1000\,\pi t) = \text{m(t)}\cos(1000\pi t) + \text{n(t)} + 3\cos(1000\,\pi t).$$

Can we use an envelope detector to substitute the coherent detector in the figure to recover message signal $m(t)$ from $\hat{r}(t)$? State the reason if the answer is "No"; otherwise find the SNR at the output of the envelope detector.

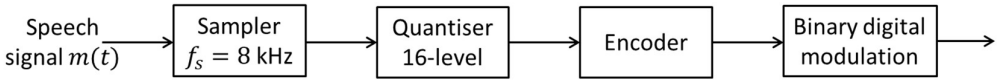
$\hat{r}(t)$ can be regarded as standard AM with

$$\mu = \frac{m_p}{A} = \frac{2}{3} \le 1.$$

Therefore, the envelope detector can be used to detect msg without distortion. In this case,

$$SNR_{AM} = \frac{P}{2N_0 W} = 166.7 \text{ or } 22.23 \text{ dB}$$

Problem IV (*Digital Communication*) The following digital communication system is used to transmit speech signal $m(t)$ with $E\{m^2(t)\} = 1$, $\max|m(t)| = 2$, and bandwidth of 3.5 kHz. Assume that sampling frequency $f_s = 8$ kHz and 16-level quantisation are used.



a) [**3 points**] Find the minimum sampling frequency in the above system to ensure accurate recovery of $m(t)$ from the sampling sequence without distortion.

$f_N = 2 \times 3.5 \text{ kHz} = 7 \text{ kHz}$

b) [**3 points**] If 16-level uniform quantisation is used, what is the SNR (due to quantisation noise) at the quantiser output?

$$SNR = \frac{3P}{m_p^2} 2^{2n} = \frac{3 \times 1}{2^2} 2^{2\times 4}$$

$$= 3 \times 2^6 \quad \text{or} \quad 22.77 \text{ dB}$$

c) [**3 points**] What is the required data transmission rate at the output of the encoder (*the encoder here just converts the 16-level discrete signal into a binary data sequence, without adding any redundancy*)?

$r_b = 8 \text{ kHz} \times 4 = 32 \text{ k bits/sec.}$

d) [**2 points**] If the data transmission rate is 50 kbits/sec and the binary digital modulation in the figure is used, what is the symbol duration and the required channel bandwidth?

$$T_b = \frac{1}{50 \times 10^3} \text{ sec} = 20 \ \mu\text{sec}$$

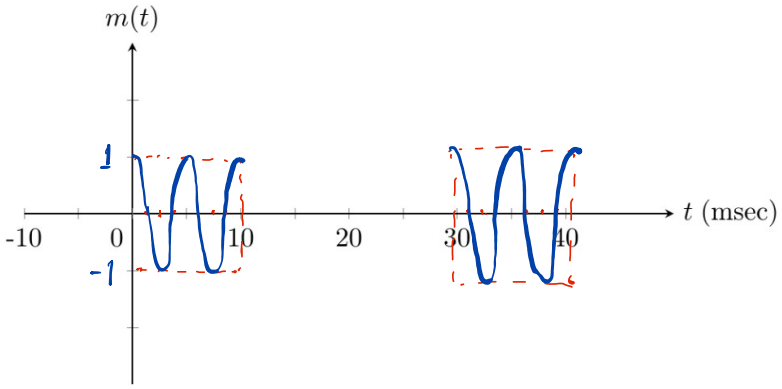
$$B = \frac{1}{2} R = \frac{1}{2} \times 50 \text{ kHz} = 25 \text{ kHz}$$

Problem V. (*Digital Modulation*) For ASK, the modulated signal can be expressed as

$$s(t) = \begin{cases} 0, & \text{if symbol "0" is transmitted,} \\ A\cos(2\pi f_c t), & \text{if symbol "1" is transmitted,} \end{cases}$$

where amplitude $A = 1$ volt, symbol duration $T = 10$ msec (0.01 sec), $f_c = 200$ Hz.

a) [**3 points**] Draw the waveform of the modulated signal if the transmitted symbol is {1,0,0,1}.



b) [**3 points**] Find the probability of error if coherent detection is used and the power spectral density (PSD) of the channel noise is $\frac{N_o}{2} = 10^{-3}$ Watt/Hz.

$\sigma^2 = N_0/T = 2\times10^{-3}/0.01 = 0.2$

$P_e = Q\left(\frac{A}{2\sqrt{\sigma^2}}\right)$

$\approx Q\left(\frac{1}{2\sqrt{0.2}}\right) = Q(1.118)$

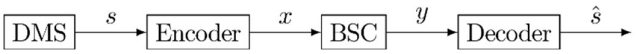
$\approx \frac{1}{\sqrt{2\pi}\cdot 1.118} e^{-\frac{1.118^2}{2}} = 0.191$

c) [**3 points**] Find the probability of error if non-coherent detection is used and the PSD of the channel noise is $\frac{N_o}{2} = 10^{-3}$ Watt/Hz.

$P_e = \frac{1}{2} e^{-A^2/8\sigma^2} = \frac{1}{2} e^{-\frac{1^2}{8\times0.2}}$

$= 0.268$

Problem VI. (*Information Theory*) In the following figure, the discrete memoryless source (DMS) randomly generates a symbol every second with $\Pr(s = a_1) = 0.4$ and $\Pr(s = a_2) = \Pr(s = a_3) = 0.3$, the binary symmetric channel (BSC) transmits two (binary) symbols every second with probability of error $P_e = 0.1$, that is, $\Pr\{y = 1|x = 0\} = \Pr\{y = 0|x = 1\} = P_e = 0.1$. The encoder in the following figure converts source symbols into binary symbols through a code scheme. (Hint: $-0.1\log_2 0.1 = 0.3322$, $-0.2\log_2 0.2 = 0.4644$, $-0.3\log_2 0.3 = 0.5211$, $-0.4\log_2 0.4 = 0.5288$, $-0.5\log_2 0.5 = 0.5000$, $-0.6\log_2 0.6 = 0.4422$, $-0.7\log_2 0.7 = 0.3602$, $-0.8\log_2 0.8 = 0.2575$, and $-0.9\log_2 0.9 = 0.1368$)



a) [**3 points**] Find the entropy of the DMS (bits/second).

$$H(S) = 0.4\log_2\frac{1}{0.4} + 2\times 0.3\log_2\frac{1}{0.3}$$
$$= 0.5288 + 2\times 0.5211$$
$$= 1.571 \text{ bits/sec.}$$

b) [**3 points**] Find the Huffman coding of the DMS and its average length (per source symbol).

	$P_i\{a_i\}$
a_1	0.4
a_2	0.3
a_3	0.3

a_2 and a_3 → 0.6 (0), a_1 → 0.4 (1); 0.6 → 1 (0)

$a_1 \to 1$, $a_2 \to 00$, $a_3 \to 01$

$\bar{l} = 1\times 0.4 + 2\times 0.3 + 2\times 0.3$

$= 1.6$ bits/sec.

c) [**3 points**] Find the capacity of the BSC (bits/second)

$$C = 1 - h(P_e) = 1 - \{0.1\log_2\frac{1}{0.1} + 0.9\log_2\frac{1}{0.9}\}$$
$$= 1 - (0.3322 + 0.1368)$$
$$= 0.531 \text{ bit/half sec.} = 1.062 \text{ bits/sec.}$$

d) [**3 points**] Are there an encoder and a decoder such that the probability of decoding error, $\Pr\{s \neq \hat{s}\}$, can be arbitrarily small? State the reason if the answer is "No"; otherwise design the encoder if the answer is "yes".

Since $C < H(S)$, no error-free encoder exists in this case.